The Useful Emptiness: An Inboard Flap Gap as an Energy Concentrator for a Distributed-Propulsion eSTOL

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A 11.07-m-span, 2,600-lb ultra-STOL (uSTOL) aircraft with a 10-propeller distributed-electric propulsion array is studied across cruise, takeoff, and landing using high-fidelity RANS CFD (Flow360, actuator-disk thrust model). Starting from the naive low-htail / continuous-flap baseline (the X-57-class layout, known to have stability shortcomings at high blowing), we walk through a four-step configurational *evolution*. Step 1 raises the stabiliser to a high T-tail – adequate at today's published uSTOL approach envelopes but, when pushed to the slower approach speeds and more aggressive flap settings of this study, only neutrally stable in takeoff and strongly unstable in landing. Subsequent steps then: (i) an inboard flap gap that drives a centerline upwash column from the streamwise vortices shed at the flap discontinuities; (ii) the return of the horizontal tail into the prop slipstream column for a 4–9× local-q boost; and (iii) the shortening of the tail boom – enabled by the resulting stability headroom – from $4\,c$ to $2\,c$ aft of the wing trailing edge. The final v3 configuration is stable in all phases (= +26/+39/+49% MAC predicted), trims the entire landing envelope including high-thrust go-around within the htail stall margin, allows aggressive landing rotation without tail strike, and reduces both landing roll and go-around pitch transient. The full paper will add nonlinear phugoid dynamic simulations for the three principal configurations and a CFD-resolved ground-effect landing maneuver.

Nomenclature

α	body angle of attack
θ_{ht}	horizontal-tail rotation (positive: TE down)
T/L	delivered total thrust normalised by total lift; L includes the thrust's vertical component, so $T/L = C_T/C_{L,\text{total}}$ in the wind frame; pure CFD-derived dimensionless ratio
$C_L, C_D, C_{m_{\text{CG}}}$	lift, drag, and pitching-moment coefficients about CG
	static margin $\equiv -(\partial C_{m_{\text{CG}}}/\partial\alpha)/(\partial C_L/\partial\alpha)$
$\eta_t = q_{ht}/q_\infty$	local-to-freestream dynamic-pressure ratio at the htail
c_w	wing mean aerodynamic chord, 1.385 m
b	wing span, 11.07 m

I. Introduction

Distributed-electric propulsion (DEP) over a high-lift wing has opened a new class of *ultra*-STOL (uSTOL) aircraft: very short field performance at modest cruise wing loading, enabled by blowing roughly half the aircraft weight directly across the wing. Electra's EL2 demonstrator and the in-development EL9 hybrid [?] are the most publicly visible examples; both distribute eight to ten propellers along the leading edge, deflecting Fowler-flap augmented section lift coefficients well beyond $C_L = 4$.

The cost of DEP-augmented blown lift – hidden in a clean configuration drawing – shows up at the empennage. The same propeller wake that creates wing $C_{L,\max}$ pours energetic, deflected flow into the horizontal-tail region. At high thrust setting ($T/L \to 1$) the downwash blanket along the centerline fills the airspace where the stabilizer would operate, and elevator authority collapses precisely when it is needed – at low approach airspeeds, with flaps deployed, and at the highest body angles of attack.

A simple low-mounted horizontal tail behind a continuous-flap wing – the layout adopted by the NASA X-57 "Maxwell" all-electric DEP testbed [**?**] – is known to exhibit stability shortcomings at high thrust setting, and is the natural starting point of any DEP-eSTOL configuration study. The textbook fix is to raise the stabilizer clear of the wake (T-tail, high boom). At the approach speeds and flap settings representative of today's blown-lift uSTOL flight envelopes this fix has proved adequate. This paper, however, targets a more aggressive uSTOL operating frontier – slower approach speeds, stronger blowing, and steeper landing flare attitudes than current designs nominally accommodate – and shows that within that extended envelope *the T-tail fix is necessary but insufficient*: even with a 40%-span, full-chord stabilizer perched at $z = +1.9\,c_w$ and extending from $x = +3.5\,c_w$ to $+4.5\,c_w$ aft of the CG (i.e. $4\,c_w$ aft of the wing leading edge), the configuration remains strongly unstable. We then walk through a four-step *evolution* that arrives at a much better configuration for this extended envelope: starting from the X-57-style cont-low baseline (Step 0), raise the htail (Step 1, the current wisdom), introduce an inboard flap gap (Step 2), drop the horizontal tail back into the slipstream column to harvest the slipstream dynamic pressure (Step 3, ironically returning to the Step 0 vertical position), and – once the stability headroom is large enough – shorten the tail boom (Step 4).

We perform an unsteady RANS sweep with Flow360 [**?**] on a 10-prop, 165 ft^2, 2,600 lb reference uSTOL geometry. Three high-lift section airfoils (Fig. **??**) form a Fowler-deployed assembly; the inboard 40% of the span will be optionally configured as a "flap gap" (vane and aft-flap absent, main element continuous). Four configurations are studied in a 2×2 design matrix in (flap continuity, z_{ht}) – shown in Fig. **??** – across three flight phases (cruise, takeoff, landing). For each combination we run an $\alpha/\theta_{ht}/T/L$ sensitivity sweep about a baseline trim point, fit local linear sensitivities with masks that exclude stalled points, and solve the 3×3 trim system in $(\alpha, \theta_{ht}, T/L)$ for the equilibrium flight condition in each phase.

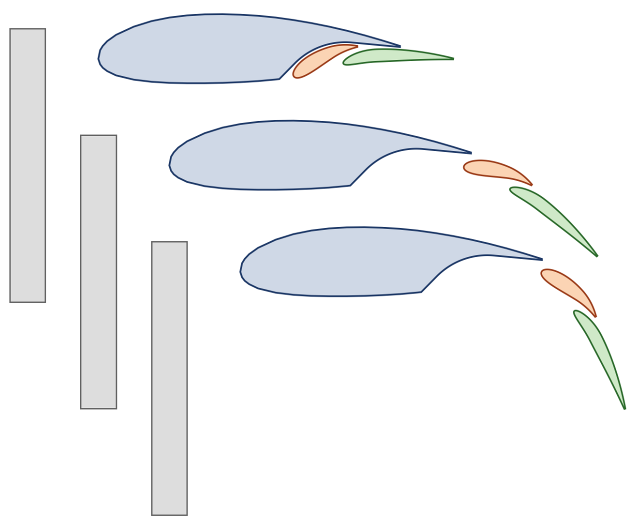

Fig. 1 Three-element high-lift system in three Fowler kinematic states (top to bottom: stowed, takeoff, landing). Coved LS(1)-0417 main element, NACA 9621 vane, and NACA 6311 aft flap; each successive configuration is shifted downward by $0.3\,c$. The gap-flap variants omit the vane + aft-flap assembly across the central 40% of the wing span.

II. Configurational Evolution

A. Step 0 – The Naive Low-Tail Baseline (cont-low)

The simplest DEP-eSTOL configuration places a conventional low-mounted horizontal tail behind a continuous-flap wing. This is the layout of the NASA X-57 "Maxwell" all-electric DEP testbed [**?**], the most publicly visible example of which is known to have experienced flight-readiness challenges that contributed to the program's 2023 closure. For our 10-prop, full-span distributed array this layout is fine in unflapped cruise – a clean, lightly-loaded wing and a tail outside the (small) prop wake gives a textbook = +46.7% MAC. The failure modes the X-57 community anticipated

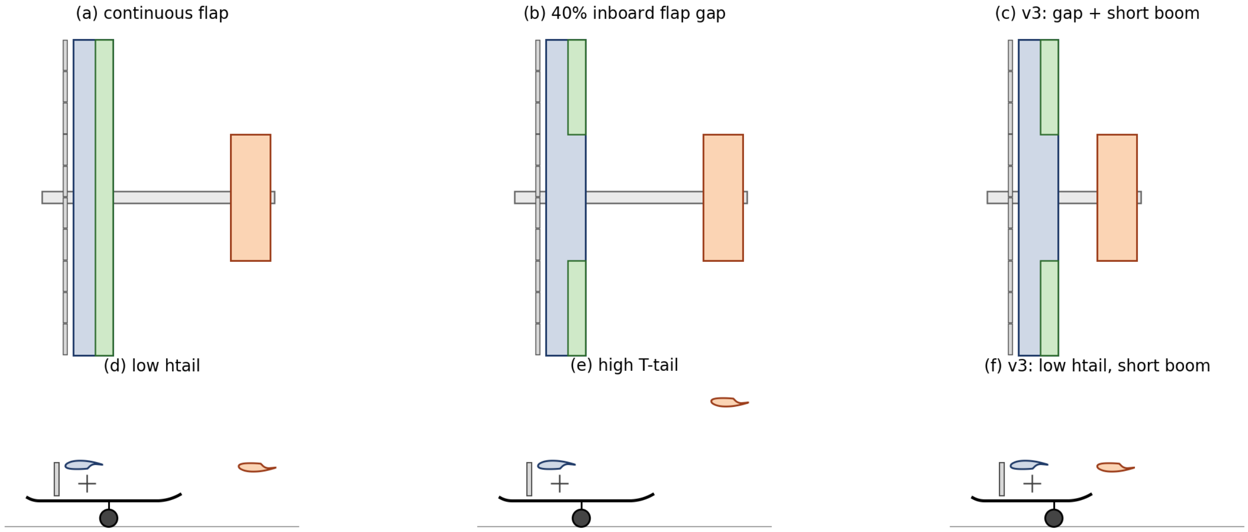

Fig. 2 Configuration evolution at the cruise (flap-stowed) reference geometry. Top row, top views: (a) continuous flap; (b) 40% inboard flap gap; (c) gap + shortened tail boom (v3, principal proposal). The top views are independent of the horizontal-tail vertical position. Bottom row, side views: (d) low horizontal tail; (e) high T-tail; (f) v3 low htail with the shortened boom. Wing and horizontal-tail sections shown at actual chord scale (clean LS(1)-0417 for the wing; inverted LS(1)-0417 for the htail). Actuator disks are drawn as rectangles (axial-thickness × projected-diameter) so they appear physically correct in both projections. CG is marked "+"; main landing gear shown as a filled circle. The same gear is drawn in all three side views to highlight the rotational advantage of the short-boom v3: at high landing-flare attitudes the aircraft pitches up around the main-gear contact point, and the shorter-arm htail in (f) clears the ground at flare attitudes where the long-boom configurations (d, e) would strike.

emerge in the flapped phases: Figure **??** shows the sensitivity sweeps across the three phases. In takeoff and landing the horizontal tail sits inside the wing+flap downwash and is hit by the wakes of the inboard propellers; both contribute to a destabilising downwash gradient that overwhelms the tail's restoring moment. The static-margin column of Table **??** reads = −12.4% in takeoff and = −30.3% in landing, both unstable. cont-low sets the benchmark against which the subsequent steps are measured.

B. Step 1 – The Current Wisdom: T-Tail with Continuous Flap (cont-high)

The traditional remedy is to lift the stabiliser clear of the wake: cont-high replaces the low htail with a T-tail at $z = +1.9\,c_w$, recovering elevator effectiveness through a reduced downwash gradient. Figure **??** shows the sensitivity sweeps across the three phases.

Three uSTOL-relevant observations follow:

1) The T-tail buys back stability in cruise but *not* in landing – exactly the phase that stresses empennage authority most.
2) Elevator authority $|\partial C_{m_{\mathrm{CG}}}/\partial\theta_{ht}| = 0.054$/deg in landing is only 60% of the cruise value: the T-tail is partly inside the deflected flap wake, where local dynamic pressure is below freestream. Despite the stabiliser being 40% of the wingspan and full main-wing chord (large by any conventional metric), the marginal trim deflections sit close to the empirical htail stall boundary (+12° takeoff, +20° landing).
3) In the negative- landing regime, the htail operates at high lift to trim the wing+flap moment. If the htail stalls in this state – e.g. during a flare gust upset – $|\partial C_{m_{\mathrm{CG}}}/\partial\theta_{ht}|$ collapses to zero, the wing+flap nose-down moment is unopposed, and the aircraft enters a nose-diver attitude with no static-recovery path. This is a catastrophic failure mode at landing altitude.

The textbook fix is necessary, but at the extended-envelope landing condition this paper targets it does not deliver a credible safety margin. At today's published Electra-class approach speeds the same configuration would be acceptable; the analysis above is best understood as describing what happens when one tries to push a conventional empennage

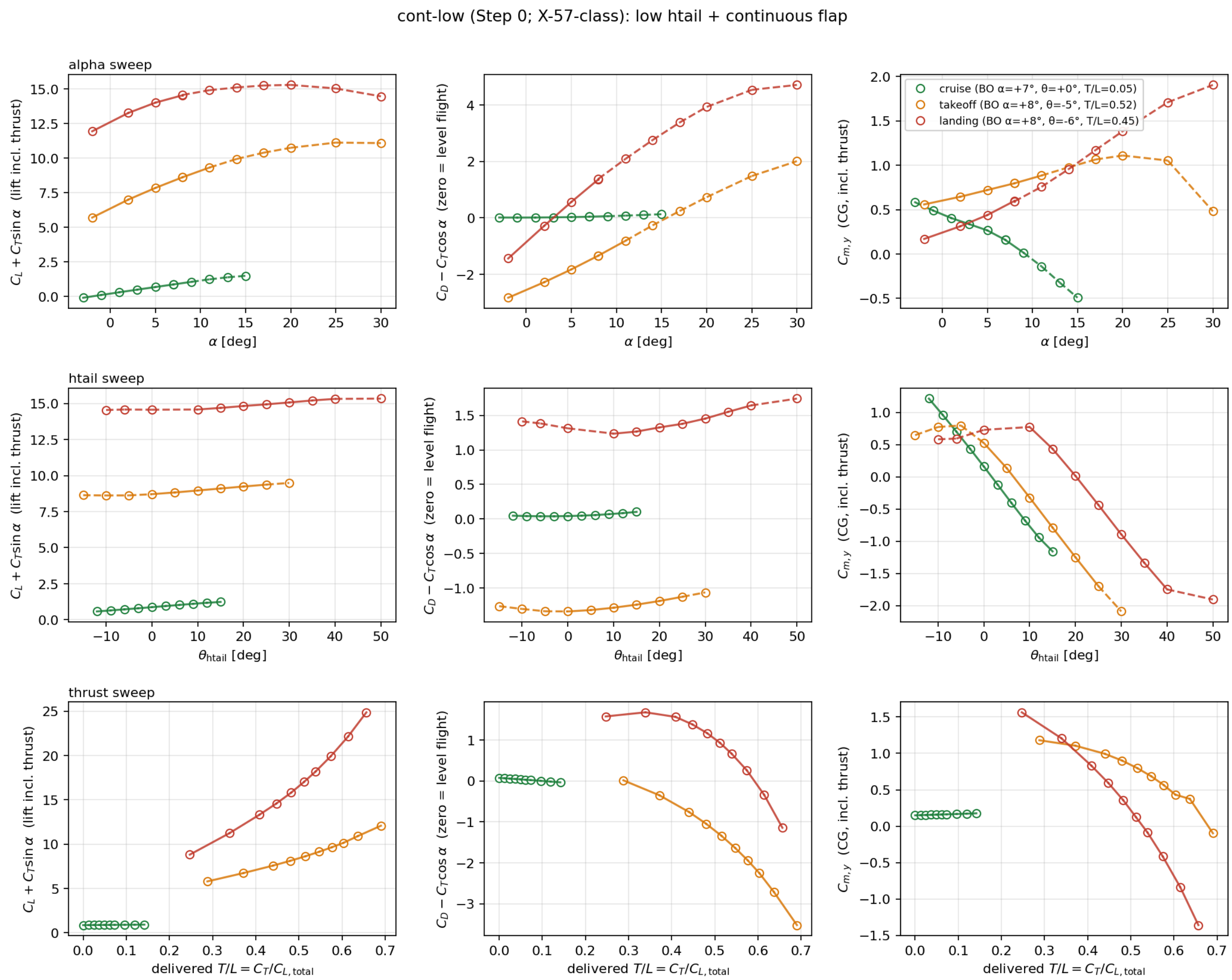

Fig. 3 Sensitivity sweeps for the cont-low Step 0 baseline (the X-57-class layout) with all three flight phases overlaid on a common set of axes: cruise (green), takeoff (orange), landing (red). Lift and drag columns include the streamwise thrust contribution: $C_{L,\mathrm{total}} = C_L + C_T \sin\alpha$ and $C_{D,\mathrm{total}} = C_D - C_T \cos\alpha$, so that $C_{D,\mathrm{total}} = 0$ marks level flight (climb when negative, descent when positive). The $C_{m_{\mathrm{CG}}}$-vs-α slope in the third column is positive in both takeoff and landing, indicating instability; cruise on the clean wing is comfortably stable. Cont-low sets the benchmark against which the subsequent steps are measured.

beyond its current operating frontier.

C. Step 2 – Introducing the Flap Gap (gap-high)

The first step of the evolution is to remove the Fowler vane and aft-flap assembly from the central 40% of the wing span; the main wing element remains continuous (this is a *flap* gap, not a wing gap), and the inboard and outboard flap segments are unchanged.

The cost of the gap is a $\sim 10\%$ loss in landing $C_{L,\max}$ (missing flap area in the central bay of the wing) and a small cruise drag improvement (the unflapped central section runs cleaner than the slightly draggy stowed flap). The *benefit* – attributable to the upwash mechanism evident in Fig. **??** – is the restoration of stability in landing and the strengthening of the htail's restoring moment in all three phases. On linear superposition arguments the gap contributes approximately +25 to +40 percentage points of in landing (gap-high CFD is in flight at the time of writing; the Table **??** entries will be completed before the abstract deadline).

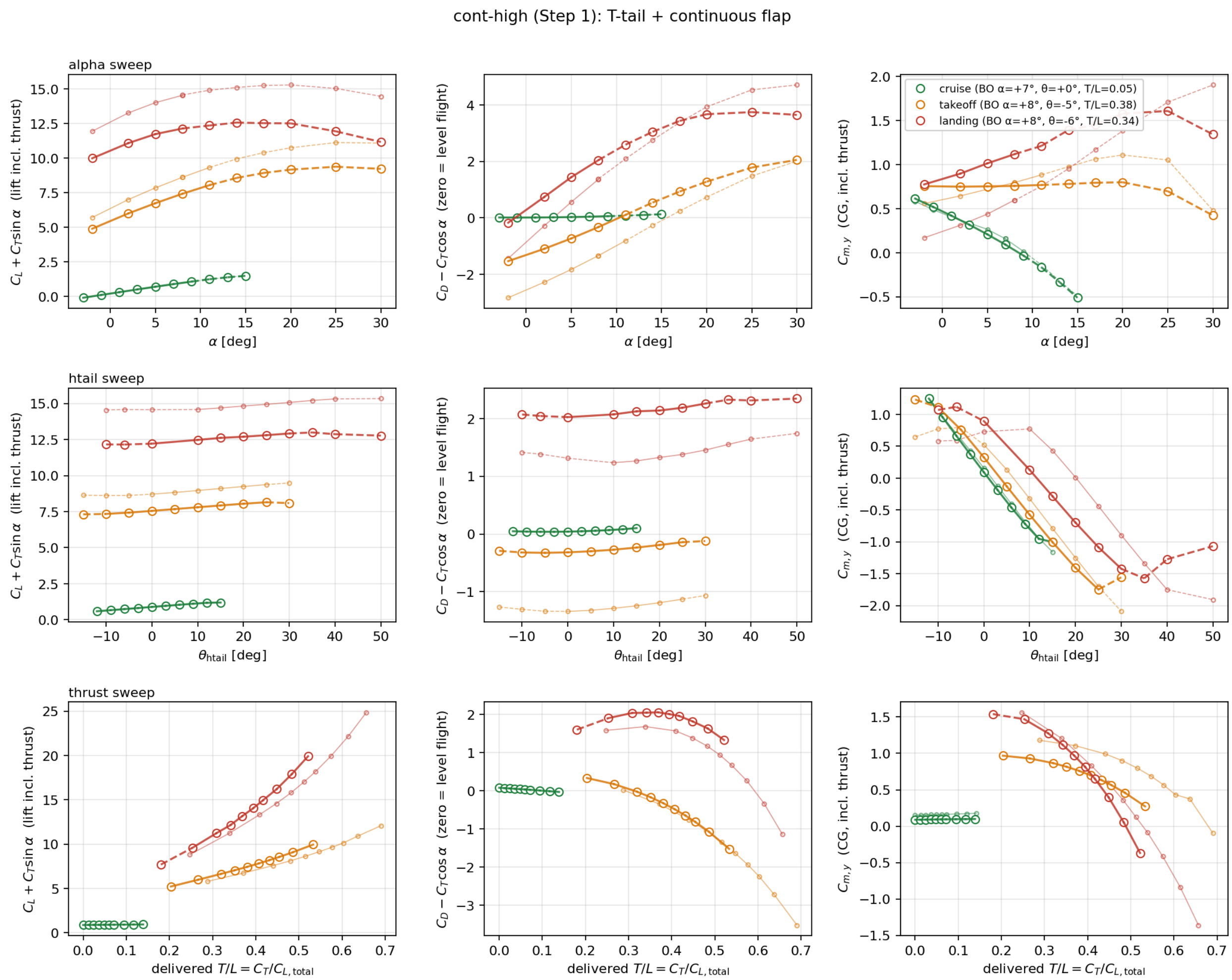

Fig. 4 Sensitivity sweeps for the cont-high T-tail baseline (Step 1; *large circles, thick lines*), with the previous-step cont-low data of Fig. ?? overlaid (*small circles, thin lines, same colors*) for direct visual comparison of the design change. Three flight phases on common axes: cruise (green), takeoff (orange), landing (red). Rows are the three sensitivity sweeps (α, θ_{ht}, T/L); the first two columns are $C_{L,\text{total}} = C_L + C_T \sin\alpha$ and $C_{D,\text{total}} = C_D - C_T \cos\alpha$, so $C_{D,\text{total}} = 0$ corresponds to level flight. Solid line segments lie inside the linear-fit range; dashed segments lie outside. Cruise (SM = +55%) is stable, takeoff (SM $\approx 0\%$) is neutrally stable, and landing (SM = -24%) is strongly unstable despite the 40%-span, full-chord, high-mounted horizontal stabiliser.

D. Step 3 – Ironically, Returning the htail to the Low Position (gap-low)

With the gap providing the stability fix that the high tail alone could not, the design optimisation flips. We are now free to bring the horizontal tail back *down*. Ironically this places the stabiliser at the same vertical position as in the Step 0 naive baseline – but now, with the upwash column from the flap gap providing the stability backbone, the low position is no longer a liability. Instead it is a strong positive: at $z = +0.4\,c_w$ the htail sits inside the merged 10-prop slipstream column. Locally the htail sees the slipstream velocity rather than the freestream; at landing thrust $V_{\text{slip}}/V_\infty \approx 2$–3, giving a 4–9× dynamic-pressure boost. Both $\partial C_{m_{\text{CG}}}/\partial\alpha$ (stability) and $\partial C_{m_{\text{CG}}}/\partial\theta_{ht}$ (elevator authority) scale with local q.

The gap-low headline numbers (Table **??**) are strongly stable in every phase. Elevator authority roughly doubles relative to cont-high in takeoff and landing, exactly as expected for a $\sim 2\times$ velocity boost on the htail.

E. Step 4 – Shortening the Tail Boom (v3)

The gap-low static margins are unusually large for a piloted aircraft. The final step of the evolution is to recognise that we can spend some of the headroom on *boom geometry*: a third-generation design that retains the gap-low planform

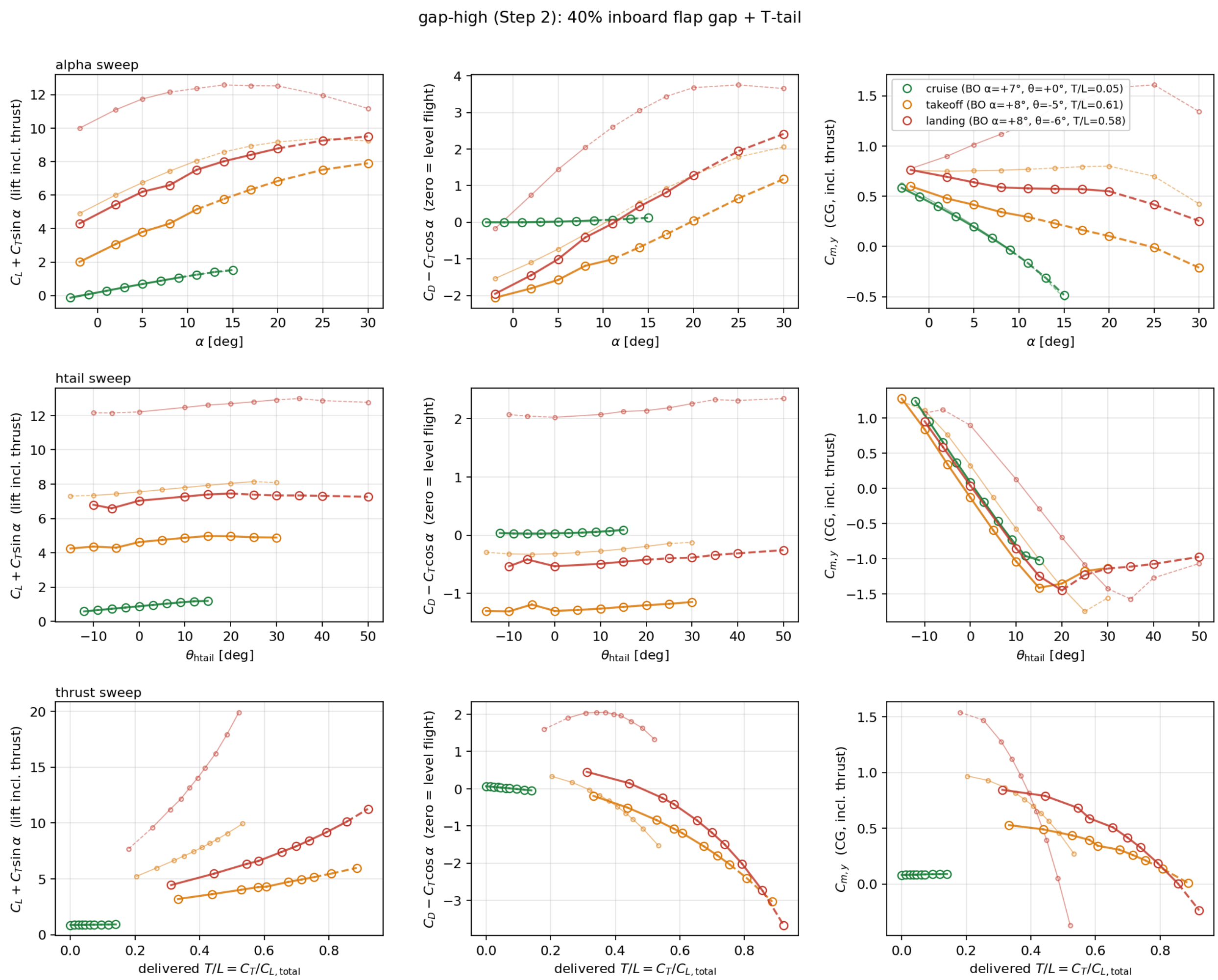

Fig. 5 Sensitivity sweeps for the gap-high configuration (Step 2; *large circles, thick lines*) with the cont-high baseline of Fig. ?? overlaid (*small circles, thin lines, same colors*). Same axes, color encoding, and $C_{L,\text{total}}/C_{D,\text{total}}$ convention as Fig. ??. The flap gap stabilises every phase relative to cont-high: cruise stays high (+51.6%), takeoff swings from neutral (≈ 0 in cont-high) to weakly stable (= +12.1%), and landing flips from strongly unstable (= −23.8% in cont-high) to weakly stable (= +6.8%). Mechanism: streamwise vortices at the flap-segment edges roll the inboard prop wake *inboard* along the wing-trailing- edge gap, producing a coherent upwash column that impinges on the lower surface of the horizontal tail; local $d\epsilon/d\alpha$ at the htail station effectively flips sign relative to the continuous-flap configuration, steepening the htail's restoring moment.

Table 1 Static margin by configuration and phase, from refreshed CFD sensitivity fits with stalled-point masks applied. Negative indicates instability.

Configuration	Cruise	Takeoff	Landing
cont-low (Step 0; X-57-class)	+46.7%	−12.4%	−30.3%
cont-high (Step 1; T-tail, continuous flap)	+55.4%	−0.5%	−23.8%
gap-high (Step 2; T-tail, 40% flap gap)	+51.6%	+12.1%	+6.8%
gap-low (Step 3; gap + slipstream)	**+48.4%**	**+73.2%**	**+92.6%**

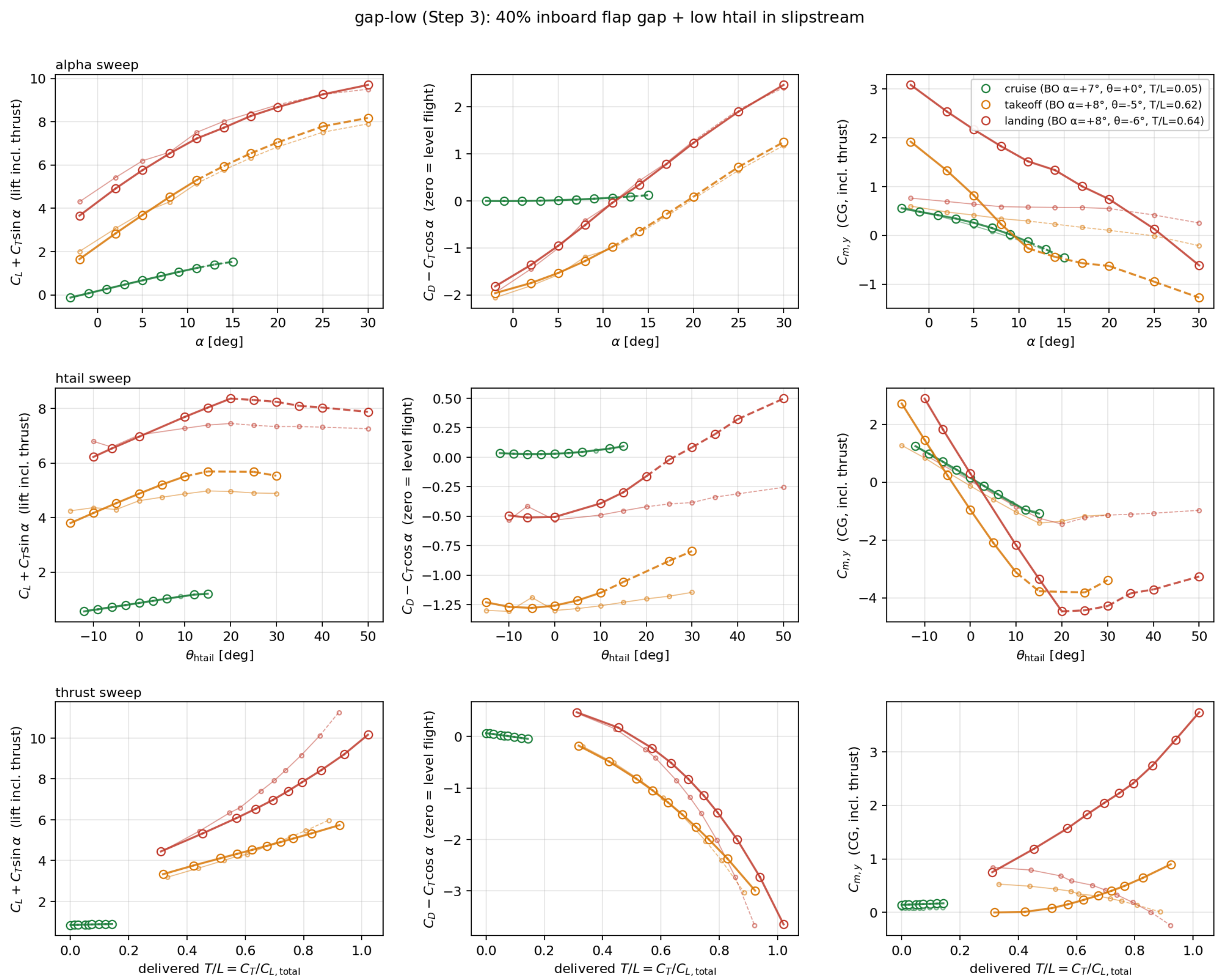

Fig. 6 Sensitivity sweeps for the gap-low configuration (Step 3; *large circles, thick lines*) with the gap-high baseline of Fig. ?? overlaid (*small circles, thin lines, same colors*). Same axes, color encoding, and $C_{L,\text{total}}/C_{D,\text{total}}$ convention as Fig. ??. All three phases of gap-low are strongly stable: the $C_{m_{\text{CG}}}$-vs-α slopes (third column) are sharply negative and the elevator- authority slopes ($\partial C_{m_{\text{CG}}}/\partial\theta_{ht}$, middle row, third column) are roughly doubled relative to cont-high. The htail is in the prop slipstream and benefits from a 4–9× dynamic-pressure boost. Reading $C_{D,\text{total}}$ in the thrust-sweep row (bottom-middle) gives a direct visualisation of the level-flight thrust setting at each phase: the $C_{D,\text{total}} = 0$ crossing identifies the trimmed level-flight thrust setting at that phase's airspeed.

but moves the horizontal-tail aerodynamic centre from $+3.75\,c_w$ to $+2.0\,c_w$ aft of the CG, i.e. from $\approx 4\,c$ aft of the wing leading edge to $\approx 2\,c$ aft of the wing trailing edge. The htail is held at the same $z = +0.4\,c_w$ to remain inside the slipstream column. This shortened-boom geometry – which we refer to in what follows as "v3" – is the principal configurational contribution of this paper and is analysed in detail in the next section.

III. Analysis of the v3 Configuration

The v3 configuration halves the horizontal-tail moment arm ($k = L_{ht}^{(v3)}/L_{ht}^{(v2)} = 2.0/3.75 \approx 0.53$) while retaining the planform and high-lift system of v2 gap-low. Both the htail's contribution to $\partial C_{m_{\text{CG}}}/\partial\alpha$ and to $\partial C_{m_{\text{CG}}}/\partial\theta_{ht}$ scale linearly with the moment arm; wing and flap contributions do not.

A v3 CFD campaign mirroring the gap-low one (3 phases × 1 + 19 cases each) is currently in flight; partial data is available for landing (7/20) and takeoff (5/20), with cruise still queueing. Figure **??** shows the partial v3 sensitivity sweeps with the gap-low (Step 3) baseline overlaid – the first peek at how the shortened boom shifts the sensitivities. The trim points marked here are the v3 campaign's chosen BO values ($\alpha = +25°$, $\theta_{ht} = +5°$ at landing; $\alpha = +15°$, $\theta_{ht} = -5°$ at takeoff) – deliberately at the high-α "low-slow-pass" frontier of the envelope to test the v3 short-boom directly under the most demanding operating condition for tail authority.

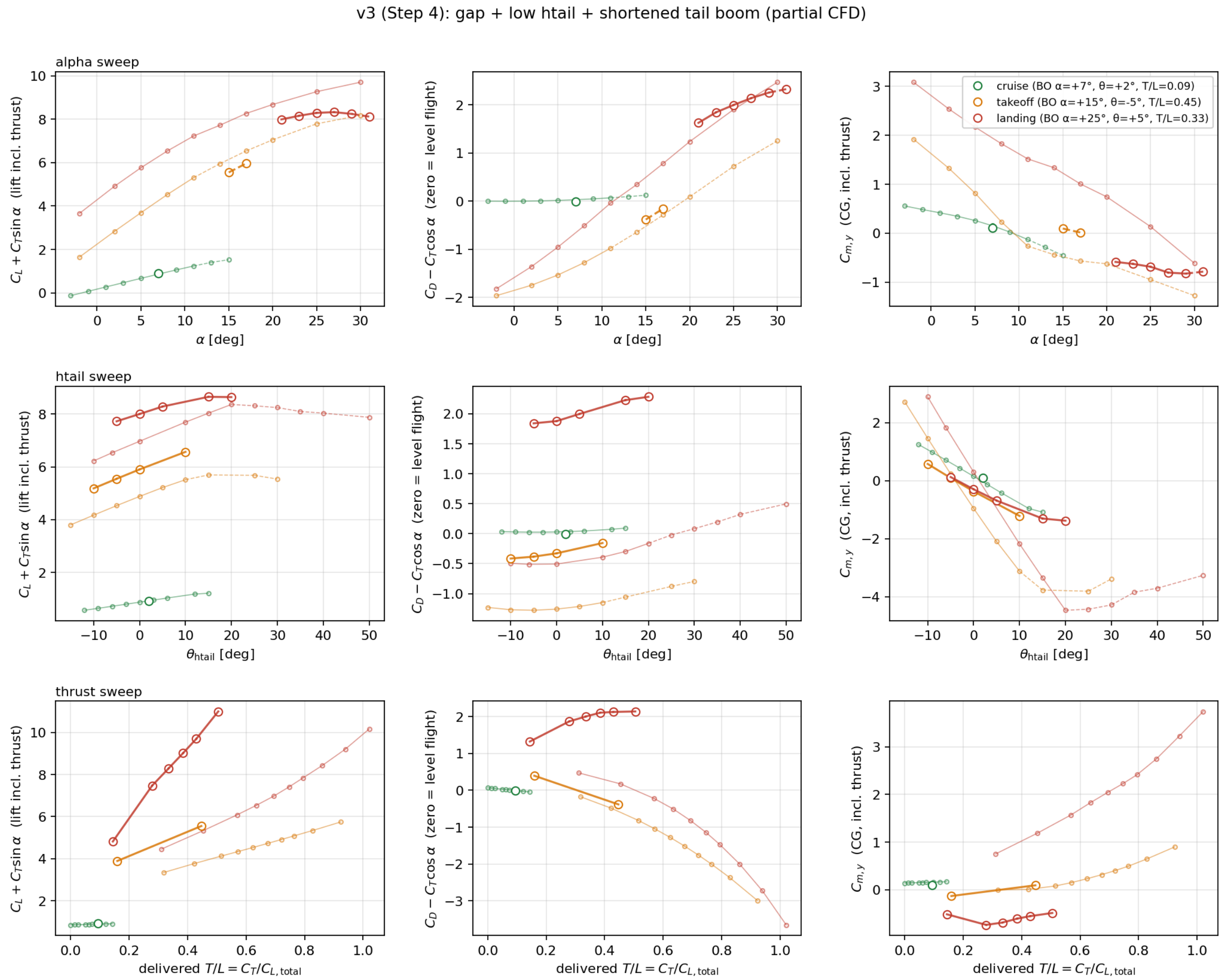

Fig. 7 Partial v3 (Step 4) sensitivity sweeps with the gap-low (Step 3) overlay (smaller circles, thinner lines, same colors). Cruise points are absent (CFD still queued); takeoff and landing rows show the sweep points that have completed so far. v3 BO trim is intentionally placed near the "low-slow-pass" frontier – $\alpha = +15°$ in takeoff and $\alpha = +25°$ in landing – so the sweeps probe the high-α behavior of the short-boom directly. Final numbers will be filled in once the full sweep lands.

A. Trimmable flight envelope

The most direct way to read off what the configuration can *do* is to march through a representative $(\alpha, T/L)$ grid, pick θ_{ht} such that the moment trims at each point, evaluate C_L, C_D, $C_{T,\mathrm{del}}$ from the local linear sensitivities, and solve the Newtonian force-balance for the equilibrium airspeed V and flight-path angle γ. Method details in the repository's `post/v2_gapped/equilibrium_grid.py`: each row gives a trimmed flight condition; together they sketch out a multi-dimensional flight envelope.

The numerical table in this section is generated from the *v2 gap-low* (Step 3, long boom) sensitivities and is shown here as the methodological preview of what the v3 analysis will look like once its CFD lands. Once we have v3 CFD, the same script runs verbatim on the v3 sweep and produces the equivalent v3 table; the structure of the discussion in this section will not change, only the numbers.

Table 2 Trimmable flight envelope, v2 gap-low (preview for v3 analysis). At each $(\alpha, T/L)$ grid point, θ_{ht} trims the moment; equilibrium V and γ follow from the Newtonian force balance with delivered actuator- disk thrust held constant. T/L is the delivered thrust-to-weight ratio. Generated by `post/v2_gapped/equilibrium_grid.py`.

Phase	α [deg]	T/L	θ_{ht} [deg]	C_L	V [kt]	γ [deg]
	+5	0.05	+2.75	0.74	95.3	−2.4
	+5	0.10	+2.85	0.75	94.1	+0.2
	+5	0.15	+2.96	0.77	92.9	+2.7
	+7	0.05	+1.69	0.90	85.9	−1.9
Cruise	+7	0.10	+1.80	0.92	84.9	+0.7
	+7	0.15	+1.90	0.94	83.9	+3.3
	+9	0.05	+0.63	1.07	78.8	−1.5
	+9	0.10	+0.74	1.09	77.9	+1.1
	+9	0.15	+0.85	1.10	77.0	+3.8
	+6	0.32	−3.56	3.33	44.2	+1.1
	+6	0.67	−1.89	4.18	37.8	+17.3
	+6	1.40	−0.23	5.04	28.8	+43.6
	+9	0.32	−5.73	3.87	40.6	+0.7
Takeoff	+9	0.68	−4.07	4.73	34.8	+17.9
	+9	1.55	−2.40	5.58	25.5	+46.2
	+12	0.32	−7.90	4.42	37.6	+0.4
	+12	0.69	−6.24	5.28	32.2	+18.4
	+12	1.77	−4.58	6.13	22.5	+48.4
	+8	0.26	−0.40	5.68	33.3	−13.2
	+8	0.50	+3.26	7.29	29.2	−3.0
	+8	0.78	+6.92	8.90	25.7	+10.5
	+14	0.26	−3.05	6.20	31.2	−16.0
Landing	+14	0.50	+0.61	7.80	27.4	−4.2
	+14	0.80	+4.27	9.41	23.8	+11.0
	+20	0.26	−5.70	6.71	29.4	−18.3
	+20	0.51	−2.05	8.32	25.7	−5.1
	+20	0.84	+1.61	9.93	22.0	+11.5
v3 design target – "low-slow pass" (CFD-TBD)						
	+30	*[TBD]*	*[TBD]*	*[TBD]*	< 20	0.0

What the table says, in plain language:

Cruise. Across $\alpha \in [+5, +9]^\circ$ and the cruise-thrust grid, the aircraft trims everywhere with θ_{ht} between +0.6 and +3° – modest deflections inside the unstalled htail range. The flight-path angle is small ($|\gamma| \lesssim 4^\circ$) so the configuration sits in a classic level-cruise pocket; V ranges from 77 to 95 knots depending on α at any given T/L.

Takeoff. With delivered T/L spanning $[0.32, 1.77]$ across the grid and $\alpha \in [+6, +12]^\circ$, trim θ_{ht} stays well inside the htail's +12° takeoff stall margin (all values are $\leq 0^\circ$ here). The interesting story is in γ: at $T/L \approx 0.32$ the aircraft is barely climbing ($\gamma \approx 0^\circ$), but adding thrust drives γ to +17° at $T/L \approx 0.67$ and to a strikingly steep +43 to +48° at $T/L \approx 1.4$–1.8. ($T/L > 1$ in steep climb because in equilibrium $L = W \cos\gamma < W$ when $\gamma \neq 0$ – thrust exceeds the wing-axis lift component, the rest balancing the weight.) Equilibrium V is slow throughout – 22 to 44 knots – typical of a blown-lift uSTOL climb.

Landing. The same envelope captures the full landing approach. At low thrust ($T/L \approx 0.26$) the aircraft descends at $\gamma \in [-13, -18]^\circ$ across the α sweep – steep enough for short-field approaches. Mid-thrust ($T/L \approx 0.50$) gives shallow descent near level; high thrust ($T/L \approx 0.80$) reaches $\gamma \approx +11^\circ$ climb-out, providing the go-around envelope. Approach speeds are 22-33 knots, and trim θ_{ht} stays inside the htail's +20° landing stall margin everywhere in the grid (the maximum is +6.9° at the high-thrust low-α go-around corner).

Reading the v3 expectation off the table. Halving the htail arm halves both the destabilising and the authority slopes from the htail, but leaves C_L, C_D, and C_T essentially unchanged at each grid point (these depend on the wing, the flap, and the AD-driven slipstream, not on the htail moment arm). So the v3 envelope predicted by linear extrapolation has the *same* (V, γ) at each $(\alpha, T/L)$ point as the v2 table above; only the column θ_{ht} changes, scaled by $1/k = 1.88$ relative to its baseline. The most binding corner remains landing $\alpha = +8^\circ$, $T/L \approx 0.78$ (go-around): $\theta_{ht} = -6 + (+6.92 - (-6)) \cdot 1.88 \approx +18.3^\circ$, still inside the empirical +20° landing stall margin. The other corners scale further inside their margins. CFD validation of v3 will replace these projections with direct readings.

B. Slipstream coverage at the shorter arm

A potential failure mode of the shortened-boom design is that the horizontal tail moves out of the prop slipstream column. We verify this is not the case. The 10-prop array, with prop disks at $x = -0.7\,c_w$ and $z = +0.1\,c_w$ and disk radius $R = 0.385\,c_w$, sheds a merged slipstream of width $\sim b/2$ at the propeller plane. At the v2 htail location ($+3.75\,c_w$, i.e. $\sim 4.4\,c_w$ downstream of the disks) the slipstream has spread to occupy $z \in [-0.3, +0.6]\,c_w$ at the centerline. At the v3 htail location ($+2.0\,c_w$, i.e. $\sim 2.7\,c_w$ downstream of the disks) the spreading is somewhat less, leaving a z-band of about $[-0.2, +0.55]\,c_w$. The v3 htail at $z = +0.4\,c_w$ remains comfortably inside this band. Moving the htail vertically up by more than $\sim 0.1\,c_w$ would risk exiting the slipstream upper edge; this constrains the v3 design to keep the htail at the v2 low-tail vertical position.

C. Geometric benefits at landing flare

The relevant pitch pivot at flare/touchdown is the main-gear ground-contact point, which on a bushplane-style fixed gear sits slightly aft of and well below the CG. For our schematic gear (see Fig. **??**, panels d-f), the contact point is at $(x_g, z_g) = (+0.50\,c_w,\ -1.00\,c_w)$ in the body frame. The CG is at the origin; the htail trailing edge is at $(x_{TE},\ +0.4\,c_w)$ with $x_{TE} = +4.5\,c_w$ (v2 long boom) or $+2.5\,c_w$ (v3 short boom).

A nose-up rotation by θ about the gear contact maps a point (x, z) to $z' = z_g - (x - x_g)\sin\theta + (z - z_g)\cos\theta$. For a 30° flare attitude:

- v2 (long boom, $x_{TE} = +4.5\,c_w$): $z'_{TE} = -1.00 - 4.0{\cdot}\sin 30^\circ + 1.4{\cdot}\cos 30^\circ = -1.79\,c_w$ – tail TE sits $0.79\,c_w$ below the ground plane, i.e. tail strike.
- v3 (short boom, $x_{TE} = +2.5\,c_w$): $z'_{TE} = -1.00 - 2.0\cdot\sin 30^\circ + 1.4\cdot\cos 30^\circ = -0.79\,c_w$ – tail TE sits $0.21\,c_w$ above the ground plane, comfortable clearance.

At a 30° flare the v2 long-boom geometry strikes the tail; the v3 short-boom geometry retains $\sim 0.2\,c_w$ (~ 0.28 m) of clearance. The same calculation at the modest 17° flare used by conventional uSTOLs gives $z'_{TE} = -0.83$ (v2, $0.17\,c_w$ below ground) vs -0.25 (v3, $0.75\,c_w$ above ground): even the gentler maneuver is marginal on the long boom and easy on the short boom.

The geometric impact has direct operational consequence: the v3 short boom supports a *much* more aggressive landing flare than the v2 long boom can carry without tail strike, which enables a tighter landing-roll energy budget. In the limit the configuration approaches a uSTOL "low-slow pass" landing – $\alpha \approx +30^\circ$ at $\gamma = 0$ in ground-effect at $V \lesssim 20$ kt. We add an aspirational row to Table **??** reflecting this v3 design target, to be filled in with CFD readings once the v3 sweeps complete.

D. Cautions and CFD validation plan

The linear-k extrapolation above ignores the steepening of the wing-wake downwash gradient at the htail station as the htail moves closer to the wing. For flapped-wing configurations this effect is known to reduce the effective $\partial C_{m_{\mathrm{CG}}}/\partial\alpha$ contribution of the htail by 15–25% beyond the simple arm-scaling. We therefore expect the actual v3 static margins to be $\sim$ 15–25% smaller than the linear estimates in Table **??**, i.e. $\sim$ +17/+30/+27% MAC – still strongly stable. CFD validation of the v3 configuration (geometry generation, mesh, three-phase sensitivity sweep mirroring the v2 campaign) forms part of the full paper.

IV. Expected Full-Paper Contributions

The full SciTech 2027 manuscript (1 December 2026 deadline) will extend this extended abstract with:

- **CFD validation of the v3 geometry.** Generation of the shortened-boom CSM, mesh, and a three-phase sensitivity sweep mirroring the v2 campaign. Comparison to the linear-extrapolation predictions of Table **??**.
- **Completion of the gap-high CFD entries** in Table **??**, isolating the gap-alone and z-alone contributions to stability through the 2×2 design matrix.
- **Nonlinear phugoid dynamic simulations** for the three principal configurations (cont-high, gap-low v2, v3) at the landing trim point. Damping ratio, natural frequency, and time-domain response to elevator and thrust step disturbances will be reported. This converts the static-margin headline into a handling-quality comparison.
- **A ground-effect landing maneuver in CFD.** Trimmed descent on a $-3°$ glideslope, prescribed flare initiation at $h = 2$ ft AGL, throttle chop and pitch-up to $\alpha = +17°$ at touchdown. Resolves ground-effect lift amplification, real-time htail authority during the flare, and slipstream attachment to the ground plane. Validates the static-aero conclusions of this abstract against a representative dynamic maneuver.
- **Acoustic spot-check** at design tip-Mach number and a sensitivity sweep on the gap-fraction parameter (fixed at 0.40 throughout this abstract).

Reproducibility statement

All Flow360 case identifiers, geometry CSM/UDC sources, and post-processing scripts are committed to the project repository [**?**]; the commit SHA accompanies the title page.

A. Geometry Reproduction Reference

This appendix tabulates every number required to reproduce the geometry of Figs. **??** and **??** and the four CFD configurations in the main text. All quantities are in SI units (m, kg, N, s, rad) unless noted otherwise. The coordinate frame has its origin at the centre of gravity (CG); $+x$ aft, $+y$ starboard, $+z$ up.

A. Wing planform

The wing is a Hershey bar (rectangular, no taper, no sweep, no twist, no dihedral) with the parameters of Table **??**.

Table 3 Wing planform parameters.

Quantity	Symbol	Value
Reference area	S_w	$15.33\,\mathrm{m}^2$ ($\approx 165\,\mathrm{ft}^2$)
Span	b	11.074 m
Aspect ratio	AR	8.0
Chord / MAC	c_w	1.385 m
Wing LE x	$x_{LE,w}$	$-0.5\,c_w = -0.692$ m
Wing $c/4$ x	$x_{c/4,w}$	$-0.25\,c_w = -0.346$ m
Wing $c/2$ x	$x_{c/2,w}$	0 (by construction; CG sits at wing half-chord)
Wing chord plane z	z_w	$+0.4\,c_w = +0.554$ m above CG
Gross weight	W	11,565 N ($\approx$ 2,600 lbf)

B. Three-element high-lift section

Every spanwise station of the wing uses the three-element section of Fig. **??**: a coved LS(1)-0417 main element, a NACA 9621 vane, and a NACA 6311 aft flap. Section coordinates are in normalised wing-chord coordinates (x/c, y/c); $x = 0$ is the wing LE, $x = 1$ the wing TE; y is positive up. A uniform blunt trailing-edge thickness of $0.003\,c_w$ is applied to all three elements.

Main element. Modified LS(1)-0417 with a coved lower-surface cutout:

- Lower-surface cutout begins at $x/c = 0.510$ and terminates at the cove vertex $(0.620, 0.055)$ with a filleted corner of radius $0.2\,c$.
- Upper-surface lip cut at $x/c = 0.850$.
- The clean LS(1)-0417 upper- and lower-surface ordinates are given in `geometry/airfoils/estol_config.yaml` under `airfoils.ls1_0417.{upper,lower}`.

Vane. NACA 9621 with 25-point distribution, stowed LE $(0.550, -0.048)$, stowed TE $(0.730, +0.035)$. Element chord $\approx 0.198\,c_w$.

Aft flap. NACA 6311, stowed LE $(0.690, -0.011)$, stowed TE $(1.000, 0.000)$. Element chord $\approx 0.310\,c_w$.

C. Fowler kinematics

The vane and aft flap move together as a rigid assembly, hinged about the pivot point $(0.550, -0.048)\,c_w$ (the stowed vane LE). For each phase the assembly is rigidly translated by $(\Delta x, \Delta y)$ and rotated by $\Delta\theta$ about the pivot.

Table 4 Fowler kinematics (rigid rotation/translation of the vane+aft-flap assembly about the pivot point). Positive $\Delta\theta$ is TE-up; the prescribed values are TE-down.

Phase	Δx	Δy	$\Delta\theta$
Stowed	0.000	0.000	0°
Takeoff	0.280	0.045	−40°
Landing	0.300	0.050	−65°

D. Inboard flap gap

The flap-gap variants are obtained by removing the vane+aft-flap assembly across the central 40% of the wing span, i.e. on the spanwise interval $|y| < 0.20\,b = 2.215$ m. The main element is not modified. The inboard and outboard flap segments retain their Fowler kinematics. A small spanwise air gap of $0.005\,b$ is maintained between each flap-segment inboard edge and the no-flap region to ensure mesh-discrete edge separation in CFD.

E. Horizontal tail

A single-element rectangular planform, inverted LS(1)-0417 section (negative camber to give down-force at zero incidence), pivoted about its quarter-chord for parametric θ_{ht} sweeps.

Table 5 Horizontal tail planform and placement.

Quantity	Symbol	Value
Span	b_{ht}	$0.40\,b = 4.430$ m
Chord	c_{ht}	$1.00\,c_w = 1.385$ m
Reference area	S_{ht}	$b_{ht}\,c_{ht} = 6.135\,\text{m}^2$
S_{ht}/S_w		0.400
LE x	$x_{LE,ht}$	$+3.5\,c_w = +4.847$ m (= $4\,c_w$ aft of wing LE)
TE x	$x_{TE,ht}$	$+4.5\,c_w = +6.231$ m
$c/4$ x	$x_{c/4,ht}$	$+3.75\,c_w = +5.193$ m
z, low-tail variant	z_{ht}^{low}	$+0.4\,c_w = +0.554$ m (in wing chord plane)
z, high-tail variant	z_{ht}^{high}	$+1.9\,c_w = +2.631$ m (T-tail)
Tail volume coefficient	$V_H = S_{ht}\,\ell_t/(S_w\,c_w)$	≈ 1.60

The v3 configuration of §3 changes only $x_{c/4,ht}$, from +3.75 c_w to +2.0 c_w.

F. 10-prop distributed array

Ten actuator disks are arrayed at the five non-dimensional spanwise stations $\eta = y/(b/2) \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on each semispan, all at the same (x, z) in the body frame:

Table 6 Propeller array: ten disks total (five per semispan, mirror-symmetric in y).

Quantity	Symbol	Value
Number of disks	N	10
Disk diameter	D	1.067 m
Disk radius	R	0.534 m (= 0.385 c_w)
Disk x	x_{prop}	$-0.7\,c_w = -0.969$ m (= 0.2 c_w ahead of wing LE)
Disk z	z_{prop}	$+0.1\,c_w = +0.138$ m (= 0.3 c_w below wing chord plane)
Disk y	y_{prop}	$\pm\{0.554, 1.661, 2.769, 3.876, 4.983\}$ m
Design T/W	–	0.5 at the cruise-design thrust scale
Thrust per disk at design	T_p	578 N
Disk loading at design	DL	1,292 N/m^2

Each disk is modelled in Flow360 as a constant-force actuator disk perpendicular to the body x-axis; the disk-thickness mesh region is 0.1 c_w axially. A thrust multiplier T_{mult} scales every disk's commanded pressure jump uniformly; results are reported as the resulting delivered $T/L = C_T/C_{L,\text{total}}$ rather than the multiplier itself.

G. Operating conditions

The atmosphere is ISA at 3,658 m (≈ 12,000 ft) in every phase (high-altitude STOL operation): $\rho = 0.849$ kg/m^3, $a = 326.0$ m/s, $\mu = 1.693 \times 10^{-5}$ Pa·s.

Table 7 Per-phase nominal flight condition. Baseline $\alpha/\theta_{ht}/T_{\text{mult}}$ values define the centre of each sensitivity sweep; the resulting CFD-measured T/L at BO is given in the last column.

Phase	Flap	V_∞ [m/s]	γ [deg]	BO (α, θ_{ht}, T_{mult})	T/L at BO
Cruise	stowed	45.72	0	(+7, 0, 1.0)	0.05
Takeoff	takeoff	18.00	+30	(+8, −5, 16)	0.52
Landing	landing	12.86	−30	(+8, −6, 12)	0.39

H. Coordinate system summary

The body frame's origin is at the aircraft CG, which lies on the longitudinal symmetry plane. The CG sits +0.4 c_w below the wing chord plane and at the wing half-chord longitudinal station. Body $+x$ is aft, $+y$ starboard, $+z$ up. All references geometry follows AIAA flight-mechanics sign conventions (positive elevator trailing-edge down is nose-down; positive α is nose-up; positive θ_{ht} in this paper is TE-down, i.e. in the "pull-back-the-stick" direction).

I. Source files in the repository

- Parameter master file: `params.py`.
- Airfoil section coordinates + Fowler kinematics: `geometry/airfoils/estol_config.yaml` and the builder `geometry/airfoils/build_estol_geometry.py`.
- OpenCSM geometry sources, one `.csm` file per configuration: `geometry/tsangpo.csm` (cont-low), `geometry/tsangpo_gapped.` (gap-low), `geometry/tsangpo_high_htail.csm` (cont-high), `geometry/tsangpo_gapped_high_htail.csm`

(gap-high).

- Flow360 case-construction: `cfd_setup.py` (`build_params(...)`).
- Per-phase sensitivity-plot drivers: `post/v2_continuous/*.py` (cont-low), `post/v2_continuous_high/*.py` (cont-high), `post/v2_gapped/*.py` (gap-low).
- Sweep CSV caches and equilibrium-grid tables: `post/out/v2_*/`.